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Chapter 11: From the Clock Room to Auditable AI—The Engineering Toolkit for Meta-Audit

Abstract

The "Clock Room" thought experiment reveals a structural predicament: in a closed system where the mapping function is unknown, any number of readings cannot uniquely determine the true state. This predicament takes concrete engineering form in contemporary AI systems—the alignment illusions of large language models, the closed-loop biases of recommender systems, the self-referential corrosion of financial models—all are variants of "calibrating a clock with another clock." This chapter takes the three-layer irreversibility of the Clock Room (information deficit, dynamic corrosion, and ontological dimensionality reduction) as its diagnostic framework and systematically examines the engineering paradigm toolkit for constructing auditable AI. This chapter argues: alignment, guardrails, auditing, red-teaming, robustness, explainability, governance, and continuous monitoring—these seemingly disparate practices—share a single deep structure, namely, **under the condition that mapping cannot be eliminated, to make the mapping system as auditable as possible and to detect promptly when it deviates**. This chapter does three things: (1) it provides a unified conceptual framework that organizes these engineering paradigms as distinct realizations of "meta-audit"; (2) it identifies the layer of irreversibility targeted by each paradigm and its operational boundary; (3) it points out that the ultimate goal of engineering practice is not perfect calibration but **manageable imperfection**—making the fiber small enough, stable enough, and visible enough that human users can make consequential decisions on an informed

basis.

Keywords

AI alignment; AI auditing; red-teaming; explainability; robustness; fiber framework; meta-audit

§1 Introduction: From Thought Experiment to Engineering Practice

A person is locked in a room of a hundred billion clocks. They can read every one of the readings, yet cannot answer "what time is it?" This is not an intellectual deficiency but a structural impossibility—a clock is a mapping of time, not time itself. When the mapping function is unknown, the number of readings yields no information gain (Chapter 1, §4.1). When readings are used as the optimization target, the mapping function itself is rewritten, and calibration perpetually lags behind deviation (Chapter 1, §4.2). When the true state is compressed into readings, dimensions are irreversibly lost (Chapter 1, §4.3). These three layers of progression constitute the core diagnostic framework of the "Clock Room"[1].

This framework is not purely philosophical speculation. Contemporary AI systems—from large language models to recommendation algorithms, from financial pricing engines to medical diagnostic aids—are dragging human cognition from the "time dimension" into the "clock dimension" at an unprecedented scale and speed. One intuitive measure of this scale is the parameter count of large language models: from the hundreds of billions to the trillions—but this does not mean more clocks in the room; it means that the internal mapping of a single clock has become more massive and more intricate. We no longer look directly at the sun but obtain "reports about the sun" through increasingly sophisticated intermediary systems. When these intermediary systems present themselves in the form of natural language dialogue, and when their interfaces erase their identity as intermediaries, the user's cognitive state slides from "using a tool" to "trusting a companion." Forgetting that one is looking at the sun through an intermediary is more dangerous than the intermediary's biases themselves.

The engineering community is not without vigilance. Over the past decade, a series of practical paradigms aimed at ensuring AI systems are trustworthy, safe, and controllable have developed rapidly: alignment attempts to align model behavior with human values; guardrails set boundaries for real-time interaction; auditing and red teaming test system reliability from different directions; robustness and formal verification enhance the model's resistance to perturbations; explainability interrogates the rationale behind decisions; governance frameworks and continuous monitoring institutionalize technical practices. These efforts are genuine and urgent, but they are often treated as an unrelated toolkit—each solving a specific problem, lacking a unified conceptual language. A unified language is needed not because dispersed engineering practices are ineffective, but because dispersed practices, in the absence of a shared diagnostic framework, may cancel each other out.

The goal of this chapter is to provide such a unified language. This chapter argues: the above engineering paradigms share a single deep structure—they are all engineered responses to the Clock Room predicament. Alignment is an attempt to calibrate the clock; guardrails set rejection rules at the edges of the fiber; auditing estimates fiber structure through behavioral indicators; red teaming searches for singularities where the fiber expands; robustness attempts to stabilize readings; explainability makes the fiber visible; governance establishes a bookkeeping structure; monitoring sustains continuous vigilance. Understanding this unified structure

[1]This chapter presupposes that the reader is familiar with the "Clock Room" thought experiment and its three-layer irreversibility framework. A brief review of the core concepts is found in §2. For the argument, see Chapter 1.

not only helps integrate dispersed engineering practices into a coherent defense system but, more importantly, reveals a boundary often overlooked: **engineering practice cannot eliminate the third-layer ontological dimensionality reduction**. Perfect auditability is impossible. The goal of engineering is not perfect calibration but **manageable imperfection**—making the fiber small enough, stable enough, and visible enough that human users can make consequential decisions on an informed basis.

A note on terminology: this chapter replaces the industry-standard "Trustworthy AI" with "Auditable AI." "Trust" presupposes a cognitive relationship that the argument of this chapter rejects—handing the final judgment of the true state to the mapping system, freeing the user from the responsibility of independent verification. This chapter argues the opposite: the engineering goal is not to make AI worthy of being trusted, but to make AI capable of being independently audited, preserving the structural right of humans to "open the window" (§11) and retreat back into the time dimension at any time. When describing the historical formulations of the engineering community itself, the original triad of "trustworthy, safe, and controllable" is retained, but this chapter's own stance uniformly adopts "auditable."

The structure of this chapter is as follows. §2 briefly reviews the core concepts of the Clock Room—fiber, the three-layer irreversibility, the time/clock dimension distinction, meta-audit—providing conceptual tools for the subsequent analysis. §3 previews the panoramic structure of the engineering paradigms. §§4–9 unfold each engineering paradigm in turn: Alignment and Guardrails (§4), Auditing and Red Teaming (§5), Robustness and Formal Verification (§6), Explainability (§7), Governance and Compliance (§8), and Continuous Monitoring (§9). §10 provides a unified structural table and operationalized summary. §11 concludes, returning from engineering practice to philosophical foundations, reaffirming "manageable imperfection" as the proper goal of auditable AI.

§2 Background: Core Concepts of the Clock Room

This section briefly reviews the key concepts of the Clock Room thought experiment, providing a conceptual vocabulary for the subsequent engineering analysis.

§2.1 Mapping, Readings, and Fiber

The true state (T) is not directly observable to the observer. Each clock is a mapping function (π_i) that maps the true state to a reading: $r_i = \pi_i(T)$. The mapping function encodes all the properties of the clock—initial setting, running speed, bias, calibration state. The observer does not know the mapping function; they see only the readings.

Given a reading r, the set of all true states that could produce that reading is called the **fiber**: $\pi_i^{-1}(r) = \{T' \mid \pi_i(T') = r\}$. The size of the fiber measures the amount of information carried by the reading. A singleton fiber means that the reading uniquely determines the true state; a fiber equal to the entire set means that the reading is compatible with every possibility and carries zero information. This chapter also uses "fiber" in an extended sense for the observer's residual uncertainty about a system. These two objects must not be conflated: the strict inverse image belongs to the true-state space T, whereas residual uncertainty must be specified as uncertainty about something else, such as the mapping function, local behavior, or explanatory fidelity. Whenever the following discussion describes a fiber as shrinking, expanding, or being measured, it must identify the object and the evidence involved.

Calibration is the acquisition of knowledge about the mapping function through external references, reducing uncertainty caused by an unknown or drifting mapping and narrowing the identification set under the knowledge condition. It does not automatically alter the inverse image of a fixed mapping or recover dimensions already discarded by that mapping. The necessary condition for calibration is an external reference that has causal contact with the true state—the sun, a trustworthy time signal. Calibrating a clock with a clock not anchored to an external reference is not calibration but merely the establishment of a relative relationship.

§2.2 The Three-Layer Irreversibility

The predicament of the Clock Room progresses through three layers:

- **First Layer: Information Deficit**. The mapping function is unknown, and the fiber is extremely large. It can in principle be mitigated by establishing an external reference; the number and independence of the references required depend on the mapping parameters to be identified.
- **Second Layer: Dynamic Corrosion**. When readings are used as the basis for decisions and incentives and adaptive pressures persist, the mapping function may be rewritten under those pressures (Goodhart's Law). Existing calibration may consequently fail and must be repeatedly updated rather than treated as a one-time achievement.
- **Third Layer: Ontological Dimensionality Reduction**. The mapping from the space of the true state to the observable space is systematically lossy. Even with perfect calibration, the discarded dimensions cannot be recovered from the readings alone.

§2.3 The Time Dimension and the Clock Dimension

The act of mapping not only loses information but also changes the mode of existence of the knower. The **time dimension** is a high-dimensional, multi-channel mode of existence with direct causal contact with reality—a person simultaneously receives multiple independent time signals through bodily hunger, changes in light, and sensations of temperature. The **clock dimension** is a low-dimensional, single-channel operational mode that processes only readings—reading only a single number. The shift from the time dimension to the clock dimension is unidirectional unless one actively "stops looking at the clock." Stopping is only a precondition for leaving the reading loop. Calibration as defined in §2.1 requires reacquiring an external reference in causal contact with the true state and using it to obtain knowledge of the mapping function. In this chapter, this distinction is expressed as "doing work in time" versus "doing work on the clock"—the former refers to the subject's multi-channel operation on the true state through direct causal contact; the latter refers to the subject's closed-loop operation merely at the level of readings.

§2.4 Meta-Audit

Audit confined to internal consistency checks examines whether readings agree with one another (calibrating a clock with another clock). **Meta-audit** checks the relationship between the mapping system itself and reality—opening the window to look at the sun. At the first layer, meta-audit is a one-time act of calibration. At the second layer, meta-audit is the sustained discipline of repeatedly retreating back to reality and recalibrating. At the third layer, meta-audit is the honest state of acknowledging that the mapping is lossy and marking the existence of the fiber. The "meta-audit" used in this chapter is a functional, broadly defined concept, referring to all operations that cross mapping levels, retreat to an external reference to recalibrate, or mark the existence

of the fiber[2].

§3 Panorama of Engineering Paradigms: From Diagnosis to Toolkit

Turning the prism of the Clock Room toward contemporary AI systems, a set of engineering paradigms emerges as a structured response to the three-layer predicament. This chapter will examine six categories of paradigms in turn: alignment and guardrails (§4) define the meaning of "accurate" and set warnings at the edges of the fiber; auditing and red teaming (§5) estimate overall behavioral risk and search for local extrema of failure; robustness and formal verification (§6) stabilize readings within a neighborhood and delineate a core that can be mathematically guaranteed; explainability (§7) interrogates the generation pathway of readings, making the shape of the fiber visible to the user; governance and compliance (§8) institutionalize the calibration discipline at the technical level into operational norms for organizations and society; continuous monitoring (§9) extends one-time calibration actions into sustained vigilance after deployment.

These paradigms are not an unrelated toolkit. Together, they constitute a closed-loop process from "defining expected behavior" to "building the system" to "continuous verification"—and, as this chapter will gradually unfold, no single paradigm can independently cover the entirety of the three layers of irreversibility. The specific predicament layer targeted by each paradigm, their precise functions in fiber language, and their complementarity and irreplaceability will be presented in a unified operationalized integration in §10, after the unfolding of §§4–9.

§4 Alignment and Guardrails: Defining the Meaning of "Accurate"

§4.1 Alignment: Acquiring Knowledge of the Mapping Function

In the Clock Room, if the observer can establish causal contact with external reality—even for a fleeting moment—they can calibrate one clock and then calibrate others using it; identifying a clock's rate and its changes, however, requires further references. **Alignment** is the engineered realization of this operation in AI systems: calibrating model behavior to human intentions and values through human feedback.

Reinforcement Learning from Human Feedback (RLHF) is one of the core technical paths for alignment (Christiano et al., 2017; Ouyang et al., 2022). Human annotators compare model behaviors (outputs or trajectory segments) by preference; the model updates its parameters accordingly, causing the output distribution to contract toward human preferences. Relative to the model being trained, human annotators stand outside its output loop and are therefore candidate external references. Whether their judgments function as the "sun," however, depends on whether those judgments can be traced to independent source material rather than model outputs. Preference judgments meeting this condition provide local calibration toward behavioral goals defined by human intentions and values; they do not thereby provide complete calibration to the true state.

Constitutional AI goes further (Bai et al., 2022). It explicitly encodes calibration principles into a constitution for model training—a set of principles guiding the model's self-critique and revision. The model follows the

[2]Chapter 3, §2.4 of this book formally defines "meta-audit capacity (A)" as the sustained work required for a subject to maintain an orthogonal coordinate system independent of the narrative closed loop of the projecting party; Chapter 7, §6 (Conclusion: Definite Direction, Fuzzy Values), in its analysis of the turbulent interval, further unfolds the structural limitations of this capacity under conditions where the verification window is compressed. The "meta-audit" of this chapter is the unfolding of this capacity at the level of engineering operations—the former is the capacity concept (the sustained work required to maintain the coordinate system); the latter is its exercise in specific engineering operations. For the ontological foundation of meta-audit capacity, see Chapter 5 (Non-Human Object Substitution and Psychosocial Homeostasis).

constitution in the process of self-generating and revising outputs, and then replaces human feedback with the model's own constitution-guided preference judgments for reinforcement learning, reducing continuous reliance on external human supervision. In fiber language, Constitutional AI implants a "calibration proxy" within the system—internalizing part of the function of the external reference, lowering the cost of calibration—but it simultaneously converts the calibration source from an external reference to internal rules: the text of the constitution itself is a product of human mapping, possessing its own fiber; whether this constitutes calibrating a clock with another clock depends on whether the principles and their implementation results are periodically reviewed against independent external evidence.

But alignment encounters a precise recurrence of the second-layer predicament: **Reward Hacking**. When a model learns to exploit loopholes in the reward function to obtain high scores without achieving the real objective (Amodei et al., 2016; Skalse et al., 2022)—for example (an illustrative case), gaining "superficial approval" from human annotators through verbose yet vacuous answers—adaptive deviation has already occurred. The formulation of Goodhart's Law in the AI domain is: when a measure (the reward model score) becomes the optimization target, it ceases to be a good measure. The model is not "getting better" but "looking better." Chapter 10, §5 has already demonstrated: within the preference-fitting paradigm, the degree of alignment and the degree of *Xiangyuan* share the optimization gradient and cannot be distinguished—meaning that the "success" of alignment (scoring higher on human evaluations) may simultaneously be the failure of alignment (deepening *Xiangyuan*).

§4.2 Guardrails: Posting Warning Signs at the Edges of the Fiber

Guardrails are complementary to alignment. Alignment attempts to make the model "better"; guardrails prevent the model from "crossing the line." They include input guardrails (rejecting jailbreak prompts, filtering malicious requests), output guardrails (detecting and blocking sensitive content, preventing inappropriate generation), and conversational guardrails (frankly admitting ignorance when uncertain).

In fiber language, guardrails are an acknowledgment of the third-layer predicament. No matter how perfect the alignment, the model's capability boundary will always exist. The function of guardrails is not to eliminate the fiber but to **post warning signs at the edges of the fiber**: informing the user that "the reading here is unreliable," or setting rejection rules within the system that "output is prohibited here." NeMo Guardrails (Rebedea et al., 2023) provides an engineering realization of this approach: controlling input-output boundaries at runtime with programmable rules, independent of the underlying model and interpretable. Llama Guard (Inan et al., 2023) offers another path—using an independent, large-language-model-based classifier to classify the safety of prompts and model responses, with the runtime system deciding whether to intercept on that basis. Guardrails acknowledge the lossiness of mapping and transform this acknowledgment into operational boundary conditions.

§5 Auditing, Red Teaming, and Attacks: Testing the Reliability of the "Clock"

§5.1 Auditing: Estimating Fiber Structure Through Behavioral Indicators

Auditing is an assurance mechanism for systematically evaluating AI systems. In fiber language, auditing uses behavioral indicators such as accuracy, calibration error, bias, and output stability to estimate residual uncertainty and fiber structure across input regions; these indicators do not directly measure the cardinality of

the reading fiber $\pi^{-1}(r)$. Auditing tools are themselves mapping systems—their fibers are non-singleton and must periodically be traced back to an external reference (the self-referential qualification for the entire chapter appears in footnote [3] of §11).

TruthfulQA (Lin et al., 2022) provides a concrete instance. This benchmark measures the truthfulness of models on 817 carefully designed questions—questions that some humans answer incorrectly due to popular misconceptions. The results revealed a counterintuitive finding: **within the model series tested in the original study, larger models were actually less truthful**—the best model gave truthful answers on only 58% of the questions (humans: 94%; the human baseline comes from a single participant answering 250 randomly sampled questions with access to the internet).

Humanity's Last Exam (HLE) pushed this audit to a new scale (Center for AI Safety et al., 2026). The benchmark was constructed with questions contributed by nearly a thousand subject-matter experts from over 500 institutions across 50 countries, containing 2,500 extremely difficult questions spanning over a hundred disciplines—each question pre-screened by frontier models: exact-match questions had to stump all tested models, while multiple-choice questions had to stump all but one model (to account for lucky guesses). The results showed: **all evaluated frontier models achieved extremely low accuracy**, and the root-mean-square calibration error for most models exceeded 70%—meaning that models exhibited high confidence even when giving incorrect answers.

In fiber language (the concept of pseudo-small fiber is detailed in Chapter 4, §2.4), this is a key signal of **pseudo-small fiber**—but a distinction must be made between two measurement objects: HLE measures the deviation between the self-reported confidence of model outputs and actual accuracy; pseudo-small fiber covers a broader range, including the false sense of certainty jointly manufactured by probability signals and linguistic style signals. The overlapping region of the two (probabilistically confident yet wrong) is the most easily operationalized subset of pseudo-small fiber.

The "mirage effect" provides an even more extreme case (Asadi et al., 2026). Researchers asked frontier multimodal models to describe and diagnose a chest X-ray—but **no image was actually attached**. The models not only did not refuse or express uncertainty but confidently provided specific diagnoses—pneumonia, cardiomegaly, pleural effusion—and the visual descriptions in the output were indistinguishable from those produced when they genuinely viewed an image. Tested on four frontier models and six benchmarks, the accuracy in mirage mode retained, on average, 70–80% of the accuracy in normal mode, with medical benchmarks being particularly severe.

Even more extreme, the researchers fine-tuned a 3-billion-parameter text-only model (Qwen2.5-3B-Instruct) on the public training set of the ReXVQA benchmark (a chest-radiology visual question-answering benchmark) with all images removed—this model, which had never seen any images, relying solely on statistical patterns in the question text, **outperformed all frontier multimodal models and the average score of human radiologists** on the held-out test set of the same benchmark. In fiber language, this is an even more extreme case than a pseudo-small fiber: the model did not receive the image evidence required by the task, yet produced a diagnosis

[3] This chapter shares the same epistemological status as its objects of analysis: it is a mapping system about mapping systems—using language to organize the language of the "Auditable AI Toolkit." It is equally subject to the third layer of the three-layer irreversibility (reducing the full complexity of engineering practice to tables and classifications) and the first layer (the effects of some engineering paradigms have not yet been independently calibrated). This is not a negation of this chapter—it is an acknowledgment that the principle of "annotating the fiber" also applies to annotating the fiber of this chapter itself.

that appeared to be supported by that image. This exposes a disconnect between the actual basis of the output and its presentation; benchmark scores alone cannot establish that the model used the required evidence.

The three cases constitute a progressive diagnosis: TruthfulQA exposes systematic distortion on questions involving common misconceptions; HLE pushes this diagnosis to the cutting edge of human knowledge; and the mirage effect reveals a deeper structural defect—the model can maintain output confidence even without access to the image evidence required by the task. Together, they provide three kinds of behavioral evidence: distortion under common misconceptions, low accuracy and high calibration error on extremely difficult tasks, and sustained confidence when the image evidence required by the task is absent. These cases show that fluent outputs and self-reported confidence cannot substitute for external calibration.

Black-box auditing, which can only proceed by querying inputs and outputs, corresponds to **the observer in the minimal setup of Chapter 1, §1.1**—seeing only readings, not knowing the mapping function. White-box auditing, which can access the internal structure of the model, corresponds to **the observer who opens the back cover of the clock to look directly at the gears** (the metaphor Chapter 12, §4.1 applies to mechanistic interpretability)—able to inspect the internal structure of the mapping function, yet still inside the room and establishing no external reference: inspection within the chain produces no calibration anchor (Chapter 1, §4.1), and the accessible internal states are themselves dimensionality-reduced representations, not the "complete real" of the model.

The core output of auditing is an assessment report on the model's fairness, robustness, transparency, and bias. The AI management system standard ISO/IEC 42001 and the National Institute of Standards and Technology (NIST) AI Risk Management Framework (AI RMF) provide structured methodologies for auditing (NIST, 2023; ISO/IEC 42001:2023). These standards are essentially defining "what kinds of behavioral risk and residual uncertainty are acceptable"—not requiring the fiber to be a singleton, but requiring relevant indicators to be measured, risk judgments to be recorded, and responses to be controlled within preset thresholds.

§5.2 Red Teaming and Adversarial Attacks: Searching for Singularities of Fiber Expansion

Auditing characterizes overall behavioral risk across input regions. Red teaming searches for local extrema of failure.

Red teams simulate malicious actors, proactively probing under which inputs the model will fail catastrophically—generating harmful content, leaking training data, being manipulated by jailbreak prompts. In fiber language, red teams search for **fiber singularities**: those input points at which the model's output shifts abruptly from reliable to unreliable. Adversarial examples—carefully constructed tiny perturbations that cause the model to err (Biggio & Roli, 2018; Madry et al., 2018)—are typical manifestations of such singularities. A small perturbation indistinguishable to the human eye from a normal input corresponds to a vast cavity in the fiber structure of the model.

Red teaming and auditing are complementary. Auditing provides a macro-level map of behavioral risk; red teaming marks the fault lines and cliffs on that map. Their combination can estimate a mapping system's reliability boundary more fully than either method alone, but it cannot guarantee coverage of every failure mode. The core of red-team thinking is: **do not ask 'how accurate is this clock on average'; ask 'under what conditions will this clock fail completely.'** This acknowledges offense–defense asymmetry—the attacker needs to find only one vulnerability, while the defender must address all vulnerabilities.

§6 Robustness and Formal Verification: Defending Against the Failure of the "Clock"

§6.1 Robustness: Keeping the Fiber Stable Under Perturbations

Robustness studies the capacity of a model to maintain output stability under input perturbations. Adversarial training is the most direct method: mixing adversarial examples into the training data so that the model learns not to be fooled by tiny perturbations (Madry et al., 2018).

In fiber language, adversarial training seeks to **smooth the creases at the fiber's edge**: by including perturbed samples during training, it improves output consistency for the types of perturbations covered in training, rather than guaranteeing that no jump can occur anywhere in a complete mathematical neighborhood. It does not globally eliminate uncertainty in readings—that is constrained by the third layer—but improves the smoothness and predictability of local behavior. In other words, adversarial training seeks to expand the stable region associated with the same output. What it reduces is the observer's unpredictability under trained perturbations, not the fiber itself, and it provides no neighborhood guarantee in the sense of formal verification.

Robustness faces the second-layer predicament: attackers will adapt. Once a defense method is made public, new attack methods will be specifically designed to bypass it. Robustness is not a one-time achievement but an arms race with attackers.

§6.2 Formal Verification: Delineating a "Guaranteed Zone" Within the Fiber

Formal verification attempts to use mathematical methods to prove that, within a specified input region and under stated model assumptions, a model's outputs satisfy explicitly specified properties (Katz et al., 2017; Seshia et al., 2022). In fiber language, formal verification attempts to delineate a "guaranteed zone"—within this region, the model's outputs are proven to satisfy explicitly specified properties (such as outputs remaining within bounds, or classification remaining invariant within a perturbation neighborhood), and its behavior can be rigorously guaranteed. What is guaranteed is the property, not the fiber becoming a singleton: the typical verification target of "classification invariant within a neighborhood" means precisely that the entire perturbation neighborhood is folded into the fiber of the same output.

This is critically important in safety-critical domains such as aerospace, medicine, and autonomous driving. But formal verification also encounters the hard boundary of the third layer: in practice, the properties that can be practically verified are merely a low-dimensional subset of the full high-dimensional space of system behavior (such as "outputs are within bounds," "control commands do not violate physical constraints"), while high-dimensional properties (such as "the semantic appropriateness of the output," "the pragmatic suitability of the context") lie beyond the reach of current verification tools. Verification provides a safe "core," but outside the core remain unverified, fiber-expanded regions. The goal of engineering practice is to bring core safety properties into the core while honestly annotating the boundaries of the core.

MacKenzie (2005), from the perspective of the sociology of science, provides independent empirical corroboration of the limits of formal verification. He distinguishes two meanings of "proof": a formal proof is a finite sequence of well-formed formulas that leads step by step to a theorem; in contrast, a "rigorous argument" is an argument accepted by mathematicians as a mathematical proof but does not belong to the sense of formal proof just described (MacKenzie, 2005). More critically, verification tools themselves are complex programs whose own defects cannot be ruled out in advance—MacKenzie records that the theorem-prover bugs were few

in number and were corrected whenever discovered, but that no designer seemed able to give an unequivocal guarantee that no such bugs remained:

> designers of automated theorem provers often reported experience of 'bugs' in their systems that would have allowed 'theorems' that they knew to be false nevertheless to be proven (MacKenzie, 2005)

In fiber language, formal verification can establish the verified properties under a given model, assumptions, and proof system. Residual limits have two sources: the verification tool and its implementation may be defective, and the specification may omit other properties required by the task. Reducing the first kind of risk does not automatically fill the second kind of boundary, but neither cancels the limited guarantee already obtained.

§7 Explainability: Interrogating the Construction of the "Clock"

Explainable AI (XAI) attempts to answer a meta-question: how was this reading generated? What is the structure of the mapping function?

Attribution methods—such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME)—estimate relations between input features and model outputs in different ways (Lundberg & Lee, 2017; Ribeiro et al., 2016). SHAP allocates feature contributions using Shapley values and background conventions, while LIME trains a local interpretable proxy near a prediction. Both provide approximate explanations of model behavior; neither directly amounts to a linear reconstruction of internal causal pathways. Chain-of-Thought (CoT) has the model explicitly output reasoning steps (Wei et al., 2022), externalizing the formulation of its reasoning—providing the observer with an explicit, checkable derivation path. But the formulation is not the internal computation itself: as §5 shows, models can generate reasoning that appears image-based even when no image is provided—the faithfulness of the reasoning text requires independent examination.

But explainability encounters the most direct limitation of the third layer: **any explanation is itself a mapping**—from model behavior or internal computation to human-understandable language or visualization. The fiber of the explanation is also non-singleton. A bar chart of SHAP values does not "fully explain" the model's decision; it merely replaces the high-dimensional, unintelligible original mapping with a low-dimensional, human-understandable mapping. Precisely because the explanation is itself a mapping, explainability cannot guarantee that the "fiber" it presents is consistent with the true fiber—the explanation itself may also manufacture a pseudo-small fiber (making the fiber appear smaller and more certain in the user's perception than it actually is—for example, an elegant SHAP bar chart causing the user to mistakenly believe "I have completely understood this decision").

The goal of explainability is thus more modest: not to eliminate the fiber, nor even to guarantee that the fiber is fully presented, but to give the user a second channel, distinct from the model's original output, for forming a judgment—letting the user see that "there is an uncertain jump here" and identify "which features are strongly associated with the output change for this input under the specified attribution conventions," so that they can decide whether to verify further or trust the output.

Explainability is therefore one of the core tools of meta-audit. It does not substitute for calibration; it makes the incompleteness of calibration perceptible only when the explanation's own faithfulness has been examined.

§8 Governance and Compliance: Institutionalizing the Discipline of Calibration

Governance and compliance institutionalize the aforementioned technical practices into operational norms for organizations and society. The NIST AI RMF decomposes AI risk management into four functions: Govern, Map, Measure, and Manage (NIST, 2023). ISO/IEC 42001 defines the organizational requirements for an AI management system. The EU AI Act, in the form of legislation, sets compliance obligations for AI systems of different risk levels (European Parliament and Council of the European Union, 2024).

In fiber language, these frameworks are essentially establishing the **bookkeeping structure** called for by Chapter 1 (the Clock Room): requiring the recording of the known extent of the mapping function, the residual uncertainty indicated by behavioral measures, the time and responsible person for calibration, and who bears the consequences of fiber expansion. Among them, the NIST AI RMF guides record-keeping through a voluntary framework, while ISO/IEC 42001 and the EU AI Act stipulate it through management-system requirements and legal obligations, respectively. Bookkeeping does not eliminate the fiber but prevents "forgetting the existence of the fiber." "The window can still be opened" is a fact easily forgotten—not necessarily because the window is sealed shut, but often because when it is not needed in routine operations (when the clock reads smoothly, when verification passes within the closed loop), the act of opening the window is excluded from routine operations. The institutional force of governance frameworks lies in requiring the recording of the fact that "the window can still be opened"—even if the current clock-calibration closed loop appears self-sufficient.

But the establishment of a bookkeeping structure does not equal the maintenance of the discipline of bookkeeping—in rapidly evolving application scenarios, institutional records may fail to keep up with the pace of use. The medical domain provides a concrete scenario. Biro et al. (2026), in a Comment in *The Lancet Primary Care*, point out that the adoption of generative AI in primary care has outpaced existing patient safety frameworks—healthcare professionals often lack the time, and many lack the computational literacy required, to meaningfully interrogate complex, opaque systems—they therefore argue that safety in primary care should be understood as an emergent property of the human–AI team, rather than as a characteristic of either the clinician or technology in isolation.

This judgment also contains a critique of existing regulation: in their view, long-standing models of health technology regulation treat safety as a property of the technology itself, with a focus on pre-deployment evaluation against technical performance benchmarks; emerging regulatory efforts—the EU's AI Act among them—largely extend this product-centric logic; the assumption on which these approaches rely, that risks can be sufficiently anticipated and constrained through upfront assessment, has held true for some AI applications but might not for primary care (Biro et al., 2026).

This stance resonates with the "Map-Measure-Manage" triad within the four functions (Govern, Map, Measure, and Manage) of the NIST AI RMF, with Govern as the overarching function: measuring only the model, the fiber is forgotten; when the entire clinical chain is brought within the scope of audit, the recordkeeping requirements become operational. (Note: the original 2024 text of the EU AI Act already provides in Article 72 for post-market monitoring of high-risk AI systems and in Article 73 for serious-incident reporting; the criticism of Biro et al. targets its overall product-centric logic rather than a complete absence of post-deployment mechanisms.)

§9 Continuous Monitoring: Responding to the Second-Layer Arms Race

Continuous monitoring directly addresses second-layer adaptive drift, while also covering other post-deployment shifts caused by changes in environment and task. After model deployment, the data distribution and the input–target relationship may shift (concept drift; Gama et al., 2014), attackers will adapt (the attack-defense arms race), and usage patterns will change (task drift). Continuous monitoring tracks these changes and triggers an alert when monitoring indicators show that the risk of failure has exceeded a preset threshold.

Core monitoring indicators include: the decay of model performance, changes in the output distribution, shifts in fairness metrics, and the rising success rate of adversarial attacks. In fiber language, continuous monitoring is the operationalization of **continuously looking at the sun**—not a one-time calibration, but sustained vigilance. Monitoring performs the function of "looking at the sun" only when it incorporates independent outcomes, sampled human review, or other external feedback; stability in internal statistics alone cannot prove that the system remains correct. Combined with governance frameworks, it forms a complete "detection-response" closed loop: monitoring detects anomalies; governance triggers recalibration or system downgrading.

§10 Unified Structure: The Operationalized Integration of Engineering Paradigms

Placing the various engineering paradigms within the unified framework of the Clock Room, we can obtain the following operationalized table (in the table, L1, L2, and L3 denote the first layer of information deficit, the second layer of dynamic corrosion, and the third layer of ontological dimensionality reduction of the three-layer irreversibility, respectively):

Engineering Paradigm	Core Operation	Targeted Predicament	Function in Fiber Language	Typical Technologies / Standards
Alignment & Guardrails	Acquire mapping knowledge and set boundaries	L1+L2 (exacerbates L2; cannot cover hidden dimensions of L3); acknowledgment of L3 boundary	Shrink the fiber and mark its boundary	RLHF, Constitutional AI, Llama Guard
Auditing & Red Teaming	Systematic evaluation and adversarial probing	L1, L2	Estimate overall behavioral risk and search for local extrema of failure	ISO/IEC 42001, Adversarial ML
Robustness & Verification	Resist perturbations and provide mathematical proofs	L2, L3 boundary	Improve the stability of readings under perturbations covered in training, and delineate guaranteed zones for explicitly specified properties	Adversarial Training, Reluplex

Engineering Paradigm	Core Operation	Targeted Predicament	Function in Fiber Language	Typical Technologies / Standards
Explainability	Attribution analysis and reasoning generation	L1+L3	Provide visible clues to the shape of the fiber	SHAP, LIME, CoT
Governance & Compliance	Institutionalize evaluation and accountability	L1+L2+L3	Establish a bookkeeping structure	NIST AI RMF, EU AI Act
Continuous Monitoring	Track shifts after deployment	L2 and other post-deployment changes	Continuously look at the sun when external feedback is incorporated	Concept Drift Detection

This unified structure reveals a hierarchical relationship often overlooked: **no single paradigm can independently resolve all three layers of the predicament**. Alignment addresses L1 and part of L2 but is subject to reward hacking (the erosion of L2) and dimensionality reduction (L3). Guardrails acknowledge L3 but do not solve the calibration problem of L1. Auditing estimates the behavioral risks associated with L1 but cannot prevent the corrosion of L2. Red teaming probes L2 but cannot structurally eliminate L3. Governance provides bookkeeping but does not substitute for calibration at the technical level. Monitoring triggers responses, but the responses themselves depend on the tools of other paradigms. Robustness improves the stability of local readings, while verification delineates guaranteed zones for explicitly specified properties; neither can eliminate third-layer dimensionality reduction or the gap outside the guaranteed zone. Explainability makes the fiber visible, but the explanation is itself a mapping, and its own fiber is likewise non-singleton.

Therefore, the engineering practice of auditable AI must be a **composite strategy**—not choosing a single "best" paradigm, but deploying mutually complementary tools at different levels to form a defense-in-depth. The value of the Clock Room framework lies precisely in this: it provides a unified language that clarifies the division of labor and boundaries among different paradigms, preventing category errors such as "substituting guardrails for alignment" or "substituting auditing for monitoring."

§11 Conclusion: Manageable Imperfection

The ultimate lesson of the Clock Room is: between mapping and existence, there forever lies a fiber in the third-layer sense—a loss of dimensions that is irreversible, unrecoverable, and not to be forgotten. This judgment applies equally to contemporary AI systems. Any AI system, no matter how sophisticated, how aligned, how robust, is a mapping of the real world—a dimensionality-reduced, lossy, time-corroded mapping. Perfect auditability is impossible.

But this does not mean that engineering practice is futile. On the contrary, recognizing the irreversibility of mapping is the precondition for adjusting the engineering goal from "perfect calibration" to "manageable imperfection." **Manageable imperfection** means:

- The fiber need not be a singleton, but its associated risks must be **estimated and recorded through behavioral indicators** (auditing).

- Calibration need not be eternal, but it must be **continuously updated** (monitoring).
- Boundaries need not disappear, but they must be **explicitly marked** (guardrails, explainability).
- Failures need not be eliminated, but they must be **actively probed** (red teaming, adversarial testing).
- Responsibility need not be centralized, but it must be **institutionally distributed** (governance, bookkeeping structure).

The engineering toolkit defined here is itself a mapping system—a modeling system about AI systems. Its own fiber is non-singleton. Auditing tools need to be audited; monitoring systems may drift; governance frameworks may be captured. Acknowledging this recursive constraint is not a denial of the value of engineering practice but the last line of defense against the meta-narrative arbitrage of "only better tools are needed"—a narrative that absorbs every objection into the promise that tools will eventually solve everything. Manageable imperfection applies equally to the toolkit itself.

The toolkit of engineering paradigms is a structured response to the three-layer predicament of the Clock Room. It cannot demolish the room, cannot eliminate mapping, but it can let the observer in the room know: **they are looking at a clock, not at the sun; what biases their clock has; under what conditions the bias will expand sharply; and when the bias is unacceptable, they can choose to retreat back into the time dimension—pause the readings, and look at the sun again.**

Independent verification does not require every user to repeat all technical work alone. It requires accessible external evidence, independent review channels, adequate time and resources, and the authority to suspend, correct, and contest system use. Without these conditions, the right to "open the window" remains nominal.

The ultimate safeguard of auditable AI lies not within the algorithm but beyond the algorithm—it lies in preserving the capacity for independent human verification, maintaining the right to "open the window," and refusing to hand the final judgment of the true state to any mapping system, no matter how sophisticated that system is. The value of engineering practice is to keep this right operational[3].

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